

3

months of data

28,018

sessions analysed

10

SQL queries written

GA4 Digital Marketing

Funnel Analysis

Google Merchandise Store

Nov 2020 – Jan 2021 | BigQuery Public Dataset

Tools: BigQuery SQL → Python → Power BI

Business Question: *"Which channels drive revenue and where is the funnel leaking?"*

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What This Analysis Covers

Following the data from traffic acquisition through to product purchase

01

Funnel Drop-off

Where are the 99.29% of visitors who don't buy dropping off?

02

Channel Performance

Which marketing channels actually drive revenue vs just traffic?

03

Traffic Quality

Sessions grew 8x in 3 months — but is that actually a good thing?

04

Audience Behaviour

New vs returning users, bounce rate, and who's actually engaging

05

Product Performance

What's selling, what's not, and where the revenue concentration risk lies

The Headlines — 3 Months of Data

Before digging in, here's what the Google Merchandise Store achieved Nov 2020 – Jan 2021

28,018

Total Sessions

198

Total Purchases

\$10,804

Total Revenue

0.71%

Conversion Rate

\$0.39

Revenue per Session

\$54.57

Avg Order Value

Only 198 out of 28,018 sessions resulted in a purchase — that's 7 in every 1,000 visitors. The 99.29% who didn't buy is where this analysis focuses.

The Funnel — 83% of Visitors Never See a Product

Session Start is excluded from chart below — the 83.2% drop to View Item makes lower stages unreadable at the same scale

Session Start: 27,901 ↓ 83.2% drop to View Item

View Item

Add to Cart

811 (2.91% of sessions)

Begin Checkout

486 (1.74% of sessions)

Payment Info

298 (1.07% of sessions)

Purchase

198 (0.71% of sessions)

Stage Drop-off Rates

Session Start → View Item

83.2%

View Item → Add to Cart

82.7%

Add to Cart → Checkout

40.1%

Checkout → Payment

38.7%

Payment → Purchase

33.6%

The real shopping funnel starts at View Item — only 16.81% of sessions get there. Once they do, the drop-offs are more moderate and actionable.

Channel Performance — Quality Beats Volume

Excluding Unknown (26,054 sessions, 0% channel coverage) and Obfuscated channels

Channel	Sessions	Purchases	Revenue	Rev/Session	CVR%
Self-Referral	1,222	113	\$5,380	\$4.4	9.2%
Google Organic	306	30	\$1,002	\$3.27	9.8%
Referral	98	9	\$2,146	\$21.9	9.2%
Direct	148	11	\$425	\$2.87	7.4%
Google Paid	26	3	\$40	\$1.54	11.5%

Internal Google ecosystem

Strongest real channel

Hidden gem — best quality

Only \$40 total — underperforming

Referral: \$21.90 / session

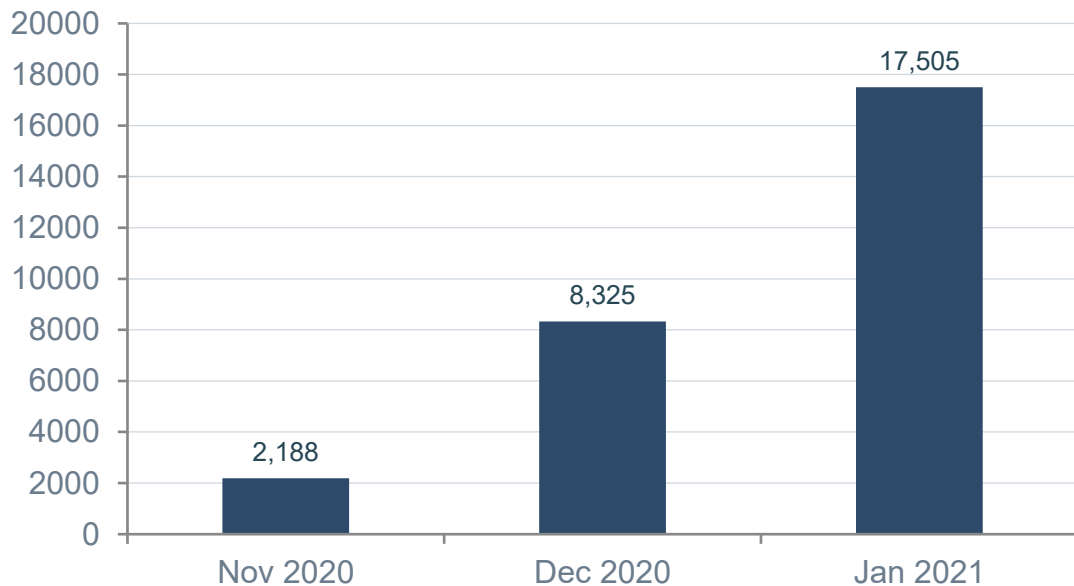
14x more valuable per visit than Google Paid. Only 98 sessions but \$2,146 revenue.

Google Paid: \$40 total revenue

11.54% CVR but only 3 purchases — statistically insignificant. Worst revenue per session at \$1.54.

Traffic Grew 8x — But Quality Fell 32%

Month-over-month sessions, CVR%, and revenue per session — Nov 2020 to Jan 2021



Monthly Sessions

Nov 2020	Dec 2020	Jan 2021
Sessions 2,188	Sessions 8,325	Sessions 17,505
CVR % 0.96%	CVR % 0.76%	CVR % 0.65%
Rev/Session \$0.68	Rev/Session \$0.64	Rev/Session \$0.23

↓ 32% CVR ↓ 66% Rev/Session

Quality declining as volume grows

January brought 17,505 sessions — 8x November. But revenue per session collapsed from \$0.68 to \$0.23. Post-holiday traffic is high-volume but low-intent. Growing sessions ≠ growing revenue.

New vs Returning — A Tale of Two Audiences

Approximated using session count per user: 1 session = New, 2+ sessions = Returning

New Users

Sessions	22,730
CVR %	0.59%
Rev / Session	\$0.29
Total Revenue	\$6,546

81% of sessions, 61% of revenue

Returning Users

Sessions	5,418
CVR %	1.18%
Rev / Session	\$0.79
Total Revenue	\$4,258

19% of sessions, 39% of revenue

Returning users are 2x more likely to purchase (1.18% vs 0.59% CVR) and generate 2.7x more revenue per session (\$0.79 vs \$0.29). Yet 81% of traffic is first-time visitors — retention is severely underinvested.

Bounce Rate — 82% of Sessions Fire Just One Event

Consistent across devices and markets, pointing to a traffic quality issue rather than a UX problem

82.29%

Overall
Bounce Rate

23,055 of
28,018 sessions

Bounce Rate by Device

Desktop	82.87%	16,388
Mobile	82.67%	11,113
Tablet	82.52%	698

Bounce rate is virtually identical across desktop (82.87%), mobile (82.67%), and tablet (82.52%). This is not a mobile UX issue — it's a traffic quality issue.

Top Markets by Sessions

Country	Sessions	Bounce%	CV R%
United States	12,419	82.59%	0.64%
India	2,686	83.88%	0.78%
Canada	2,112	84.23%	0.99%
UK	893	84.21%	0.56%
France	549	81.60%	0.73%
Germany	542	83.39%	0.55%
China	490	80.61%	0.61%
Spain	512	83.79%	1.37%

⚠ Italy: 409 sessions, \$0 revenue — investigate

Product Performance — Apparel Dominates Everything

47% of total revenue from one category is a concentration risk worth addressing

Top 5 Categories

Category	Revenue	Orders	AOV
Apparel	\$171,727	4,984	\$34.46
New	\$25,813	1,445	\$17.86
Bags	\$23,860	704	\$33.89
Campus Coll.	\$20,061	1,497	\$13.40
Accessories	\$17,815	1,276	\$13.96

Top 5 Products by Revenue

Google Zip Hoodie F/C	\$13,788
Google Crewneck Sweatshirt Navy	\$10,714
Google Men's Tech Fleece Grey	\$9,965
Google Badge Heavyweight Pullover	\$9,712
Super G Unisex Joggers	\$9,529

Apparel monthly trend: \$75,070 (Nov) → \$66,962 (Dec) → \$29,695 (Jan) — 60% revenue collapse post-holiday. Other categories remained stable.

All top 10 products are Apparel. Every channel is Apparel-dominated. Google Paid drives zero Apparel sales — missing the highest-value category entirely.

10 Things This Data Is Telling Us

A summary of the most actionable findings across funnel, channel, audience, and product analysis

1

CVR is 0.71% — only 7 in every 1,000 sessions buy. The biggest leak is at View Item (83.2% drop).

2

Referral traffic generates \$21.90 per session — 14x more than Google Paid (\$1.54). Scale it.

3

Google Paid: \$40 total revenue from 26 sessions across 3 months. Needs urgent review.

4

Sessions grew 8x but CVR fell from 0.96% to 0.65%. January traffic is low-intent noise.

5

Bounce rate is ~82% across all devices. Not a mobile UX problem — it's traffic quality.

6

Returning users: 2x CVR, 2.7x revenue per session. 81% of traffic is new users — retention underinvested.

7

Apparel = 47% of revenue and heavily seasonal. Revenue fell 60% Nov to Jan.

8

Direct channel: 59% drop at payment stage — payment friction is the biggest checkout problem.

9

Italy: 409 sessions, \$0 revenue. Either tracking is broken or shipping is blocked.

10

Bags (\$33.89 AOV) and Apparel (\$34.46 AOV) are high-value. Campus Collection is high volume, low price.

Three Things Worth Testing Right Now

Based on the data — not guesswork. Each has a clear hypothesis and measurable outcome.

A/B Test 01

Fix the View Item → Add to Cart Drop

Hypothesis

82.7% of users who view a product don't add it to cart. Hypothesis: adding urgency signals (low stock, limited time) on product pages will increase ATC rate.

Target: ATC rate from 17.3% → 22%+

Effort: Medium | Impact: High

A/B Test 02

Fix Direct Channel Payment Abandonment

Hypothesis

Direct traffic drops 59.3% at Payment Info → Purchase. Hypothesis: adding trust signals (security badges, money-back guarantee) at payment will reduce drop-off.

Target: Payment → Purchase from 40.7% → 50%+

Effort: Low | Impact: High

Strategy 03

Invest in Retention — Email/Loyalty

Hypothesis

Returning users have 2x CVR and 2.7x revenue per session. A basic email remarketing sequence targeting first-time buyers could meaningfully shift the new/returning ratio.

Target: Returning user sessions from 19% → 25% of traffic

Effort: Medium | Impact: Very High

Thank you

for taking the time to go through this.

The full project is live on GitHub: github.com/rupsa723/ga4-funnel-analysis

What's in the repo:

- 10 BigQuery SQL files — fully commented with business context
- Python data preparation script (pandas pipeline)
- Power BI dashboard (.pbix)
- Project documentation